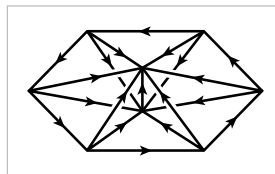


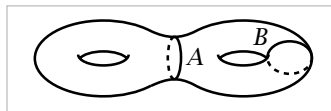
in a 1944 paper of Eilenberg, after contributions from quite a few others, particularly Alexander and Lefschetz. Within the next few years the rest of the basic structure of homology theory as we have presented it fell into place, and the first definitive treatment appeared in the classic book [Eilenberg & Steenrod 1952].

Exercises

1. What familiar space is the quotient Δ -complex of a 2-simplex $[v_0, v_1, v_2]$ obtained by identifying the edges $[v_0, v_1]$ and $[v_1, v_2]$, preserving the ordering of vertices?
2. Show that the Δ -complex obtained from Δ^3 by performing the edge identifications $[v_0, v_1] \sim [v_1, v_3]$ and $[v_0, v_2] \sim [v_2, v_3]$ deformation retracts onto a Klein bottle. Find other pairs of identifications of edges that produce Δ -complexes deformation retracting onto a torus, a 2-sphere, and $\mathbb{R}P^2$.
3. Construct a Δ -complex structure on $\mathbb{R}P^n$ as a quotient of a Δ -complex structure on S^n having vertices the two vectors of length 1 along each coordinate axis in $\mathbb{R}^{n+1}$.
4. Compute the simplicial homology groups of the triangular parachute obtained from Δ^2 by identifying its three vertices to a single point.
5. Compute the simplicial homology groups of the Klein bottle using the Δ -complex structure described at the beginning of this section.
6. Compute the simplicial homology groups of the Δ -complex obtained from $n + 1$ 2-simplices $\Delta_0^2, \dots, \Delta_n^2$ by identifying all three edges of Δ_0^2 to a single edge, and for $i > 0$ identifying the edges $[v_0, v_1]$ and $[v_1, v_2]$ of Δ_i^2 to a single edge and the edge $[v_0, v_2]$ to the edge $[v_0, v_1]$ of Δ_{i-1}^2 .
7. Find a way of identifying pairs of faces of Δ^3 to produce a Δ -complex structure on S^3 having a single 3-simplex, and compute the simplicial homology groups of this Δ -complex.
8. Construct a 3-dimensional Δ -complex X from n tetrahedra $T_1, \dots, T_n$ by the following two steps. First arrange the tetrahedra in a cyclic pattern as in the figure, so that each T_i shares a common vertical face with its two neighbors T_{i-1} and T_{i+1} , subscripts being taken mod n . Then identify the bottom face of T_i with the top face of T_{i+1} for each i . Show the simplicial homology groups of X in dimensions 0, 1, 2, 3 are $\mathbb{Z}, \mathbb{Z}_n, 0, \mathbb{Z}$, respectively. [The space X is an example of a *lens space*; see Example 2.43 for the general case.]
9. Compute the homology groups of the Δ -complex X obtained from Δ^n by identifying all faces of the same dimension. Thus X has a single k -simplex for each $k \leq n$.
10. (a) Show the quotient space of a finite collection of disjoint 2-simplices obtained by identifying pairs of edges is always a surface, locally homeomorphic to $\mathbb{R}^2$.
(b) Show the edges can always be oriented so as to define a Δ -complex structure on the quotient surface. [This is more difficult.]



11. Show that if A is a retract of X then the map $H_n(A) \rightarrow H_n(X)$ induced by the inclusion $A \subset X$ is injective.
12. Show that chain homotopy of chain maps is an equivalence relation.
13. Verify that $f \simeq g$ implies $f_* = g_*$ for induced homomorphisms of reduced homology groups.
14. Determine whether there exists a short exact sequence $0 \rightarrow \mathbb{Z}_4 \rightarrow \mathbb{Z}_8 \oplus \mathbb{Z}_2 \rightarrow \mathbb{Z}_4 \rightarrow 0$. More generally, determine which abelian groups A fit into a short exact sequence $0 \rightarrow \mathbb{Z}_{p^m} \rightarrow A \rightarrow \mathbb{Z}_{p^n} \rightarrow 0$ with p prime. What about the case of short exact sequences $0 \rightarrow \mathbb{Z} \rightarrow A \rightarrow \mathbb{Z}_n \rightarrow 0$?
15. For an exact sequence $A \rightarrow B \rightarrow C \rightarrow D \rightarrow E$ show that $C = 0$ iff the map $A \rightarrow B$ is surjective and $D \rightarrow E$ is injective. Hence for a pair of spaces (X, A) , the inclusion $A \hookrightarrow X$ induces isomorphisms on all homology groups iff $H_n(X, A) = 0$ for all n .
16. (a) Show that $H_0(X, A) = 0$ iff A meets each path-component of X .
(b) Show that $H_1(X, A) = 0$ iff $H_1(A) \rightarrow H_1(X)$ is surjective and each path-component of X contains at most one path-component of A .
17. (a) Compute the homology groups $H_n(X, A)$ when X is S^2 or $S^1 \times S^1$ and A is a finite set of points in X .
(b) Compute the groups $H_n(X, A)$ and $H_n(X, B)$ for X a closed orientable surface of genus two with A and B the circles shown. [What are X/A and X/B ?]
18. Show that for the subspace $\mathbb{Q} \subset \mathbb{R}$, the relative homology group $H_1(\mathbb{R}, \mathbb{Q})$ is free abelian and find a basis.
19. Compute the homology groups of the subspace of $I \times I$ consisting of the four boundary edges plus all points in the interior whose first coordinate is rational.
20. Show that $\tilde{H}_n(X) \approx \tilde{H}_{n+1}(SX)$ for all n , where SX is the suspension of X . More generally, thinking of SX as the union of two cones CX with their bases identified, compute the reduced homology groups of the union of n cones CX with their bases identified.
21. Making the preceding problem more concrete, construct explicit chain maps $s: C_n(X) \rightarrow C_{n+1}(SX)$ inducing isomorphisms $\tilde{H}_n(X) \rightarrow \tilde{H}_{n+1}(SX)$.
22. Prove by induction on dimension the following facts about the homology of a finite-dimensional CW complex X , using the observation that X^n/X^{n-1} is a wedge sum of n -spheres:
 - (a) If X has dimension n then $H_i(X) = 0$ for $i > n$ and $H_n(X)$ is free.
 - (b) $H_n(X)$ is free with basis in bijective correspondence with the n -cells if there are no cells of dimension $n - 1$ or $n + 1$.
 - (c) If X has k n -cells, then $H_n(X)$ is generated by at most k elements.



23. Show that the second barycentric subdivision of a Δ -complex is a simplicial complex. Namely, show that the first barycentric subdivision produces a Δ -complex with the property that each simplex has all its vertices distinct, then show that for a Δ -complex with this property, barycentric subdivision produces a simplicial complex.
24. Show that each n -simplex in the barycentric subdivision of Δ^n is defined by n inequalities $t_{i_0} \leq t_{i_1} \leq \cdots \leq t_{i_n}$ in its barycentric coordinates, where $(i_0, \dots, i_n)$ is a permutation of $(0, \dots, n)$.
25. Find an explicit, noninductive formula for the barycentric subdivision operator $S: C_n(X) \rightarrow C_n(X)$.
26. Show that $H_1(X, A)$ is not isomorphic to $\tilde{H}_1(X/A)$ if $X = [0, 1]$ and A is the sequence $1, 1/2, 1/3, \dots$ together with its limit 0. [See Example 1.25.]
27. Let $f: (X, A) \rightarrow (Y, B)$ be a map such that both $f: X \rightarrow Y$ and the restriction $f: A \rightarrow B$ are homotopy equivalences.
- (a) Show that $f_*: H_n(X, A) \rightarrow H_n(Y, B)$ is an isomorphism for all n .
- (b) For the case of the inclusion $f: (D^n, S^{n-1}) \hookrightarrow (D^n, D^n - \{0\})$, show that f is not a homotopy equivalence of pairs — there is no $g: (D^n, D^n - \{0\}) \rightarrow (D^n, S^{n-1})$ such that fg and gf are homotopic to the identity through maps of pairs. [Observe that a homotopy equivalence of pairs $(X, A) \rightarrow (Y, B)$ is also a homotopy equivalence for the pairs obtained by replacing A and B by their closures.]
28. Let X be the cone on the 1-skeleton of Δ^3 , the union of all line segments joining points in the six edges of Δ^3 to the barycenter of Δ^3 . Compute the local homology groups $H_n(X, X - \{x\})$ for all $x \in X$. Define ∂X to be the subspace of points x such that $H_n(X, X - \{x\}) = 0$ for all n , and compute the local homology groups $H_n(\partial X, \partial X - \{x\})$. Use these calculations to determine which subsets $A \subset X$ have the property that $f(A) \subset A$ for all homeomorphisms $f: X \rightarrow X$.
29. Show that $S^1 \times S^1$ and $S^1 \vee S^1 \vee S^2$ have isomorphic homology groups in all dimensions, but their universal covering spaces do not.
30. In each of the following commutative diagrams assume that all maps but one are isomorphisms. Show that the remaining map must be an isomorphism as well.

$$\begin{array}{ccc}
 A \longrightarrow B & A \longrightarrow B & A \longrightarrow B \\
 \searrow \quad \nearrow & \downarrow \quad \downarrow & \downarrow \quad \uparrow \\
 C & C \longrightarrow D & C \longrightarrow D
 \end{array}$$

31. Using the notation of the five-lemma, give an example where the maps α , β , δ , and ε are zero but γ is nonzero. This can be done with short exact sequences in which all the groups are either $\mathbb{Z}$ or 0.

2.2 Computations and Applications

Now that the basic properties of homology have been established, we can begin to move a little more freely. Our first topic, exploiting the calculation of $H_n(S^n)$, is Brouwer's notion of degree for maps $S^n \rightarrow S^n$. Historically, Brouwer's introduction of this concept in the years 1910–12 preceded the rigorous development of homology, so his definition was rather different, using the technique of simplicial approximation which we explain in §2.C. The later definition in terms of homology is certainly more elegant, though perhaps with some loss of geometric intuition. More in the spirit of Brouwer's definition is a third approach using differential topology, presented very lucidly in [Milnor 1965].

Degree

For a map $f: S^n \rightarrow S^n$ with $n > 0$, the induced map $f_*: H_n(S^n) \rightarrow H_n(S^n)$ is a homomorphism from an infinite cyclic group to itself and so must be of the form $f_*(\alpha) = d\alpha$ for some integer d depending only on f . This integer is called the **degree** of f , with the notation $\deg f$. Here are some basic properties of degree:

- (a) $\deg \mathbb{1} = 1$, since $\mathbb{1}_* = \mathbb{1}$.
- (b) $\deg f = 0$ if f is not surjective. For if we choose a point $x_0 \in S^n - f(S^n)$ then f can be factored as a composition $S^n \rightarrow S^n - \{x_0\} \hookrightarrow S^n$ and $H_n(S^n - \{x_0\}) = 0$ since $S^n - \{x_0\}$ is contractible. Hence $f_* = 0$.
- (c) If $f \simeq g$ then $\deg f = \deg g$ since $f_* = g_*$. The converse statement, that $f \simeq g$ if $\deg f = \deg g$, is a fundamental theorem of Hopf from around 1925 which we prove in Corollary 4.25.
- (d) $\deg fg = \deg f \deg g$, since $(fg)_* = f_*g_*$. As a consequence, $\deg f = \pm 1$ if f is a homotopy equivalence since $fg \simeq \mathbb{1}$ implies $\deg f \deg g = \deg \mathbb{1} = 1$.
- (e) $\deg f = -1$ if f is a reflection of S^n , fixing the points in a subsphere S^{n-1} and interchanging the two complementary hemispheres. For we can give S^n a Δ -complex structure with these two hemispheres as its two n -simplices Δ_1^n and Δ_2^n , and the n -chain $\Delta_1^n - \Delta_2^n$ represents a generator of $H_n(S^n)$ as we saw in Example 2.23, so the reflection interchanging Δ_1^n and Δ_2^n sends this generator to its negative.
- (f) The antipodal map $-\mathbb{1}: S^n \rightarrow S^n$, $x \mapsto -x$, has degree $(-1)^{n+1}$ since it is the composition of $n+1$ reflections, each changing the sign of one coordinate in $\mathbb{R}^{n+1}$.
- (g) If $f: S^n \rightarrow S^n$ has no fixed points then $\deg f = (-1)^{n+1}$. For if $f(x) \neq x$ then the line segment from $f(x)$ to $-x$, defined by $t \mapsto (1-t)f(x) - tx$ for $0 \leq t \leq 1$, does not pass through the origin. Hence if f has no fixed points, the formula $f_t(x) = [(1-t)f(x) - tx]/|(1-t)f(x) - tx|$ defines a homotopy from f to